Report

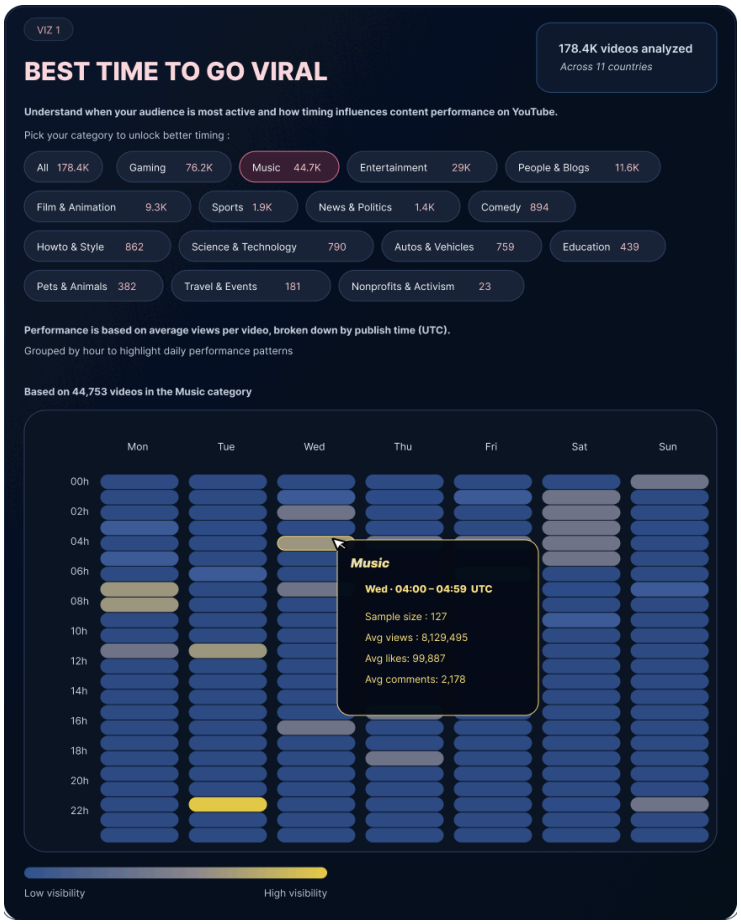
As introduced in Milestone 1, the goal of this project is to understand how YouTube creators optimize their video metadata to improve visibility on the platform.

Rather than simply analyzing what content becomes popular, we focus on the strategies behind content creation and how metadata is intentionally designed to enhance performance. To address this problematic, we analyze multiple dimensions of metadata with the aim of providing actionable insights for content creators and overall users seeking to improve their performance.

We have already developed an initial skeleton of our visualization system, which is available through our website.

For concrete insights, we designed a set of visualizations on Figma that explore different dimensions of video metadata. Each visualization focuses on a specific aspect of metadata optimization, allowing users to understand how different factors contribute to video performance.

1 . Time-Based Performance Heatmap

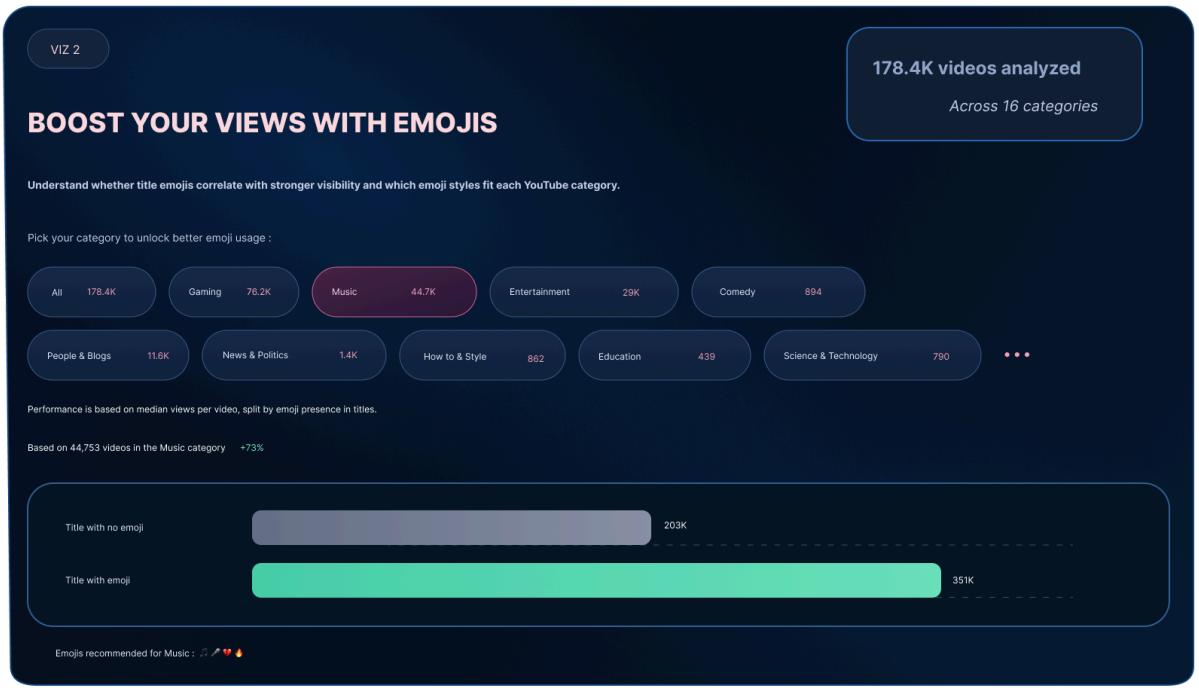


This visualization will be implemented using D3.js with SVG, using band scales to structure the grid and a sequential color scale to encode performance. Each time slot is represented by a rectangle, with tooltips and category filtering enabling interaction.

This visualization relies on the following lectures :

- *D3.js*
- *Perception, cognition, color*
- *Mark and channels*

2. Emoji Performance Bar Chart

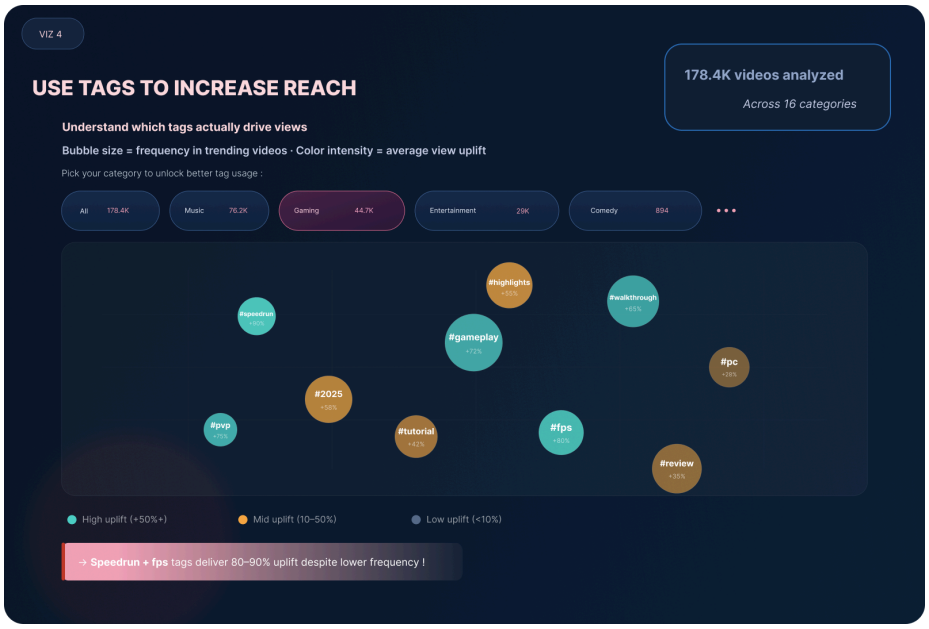


This visualization will be implemented using D3.js with SVG, with bar length encoding average views depending on emoji usage in video titles. It uses linear and band scales to structure the chart, while transitions allow smooth updates when users filter by category.

This visualization relies on the following lectures:

- D3.js
- Interaction, filtering, aggregation
- Mark and channels

3. Tag Performance Bubble Chart



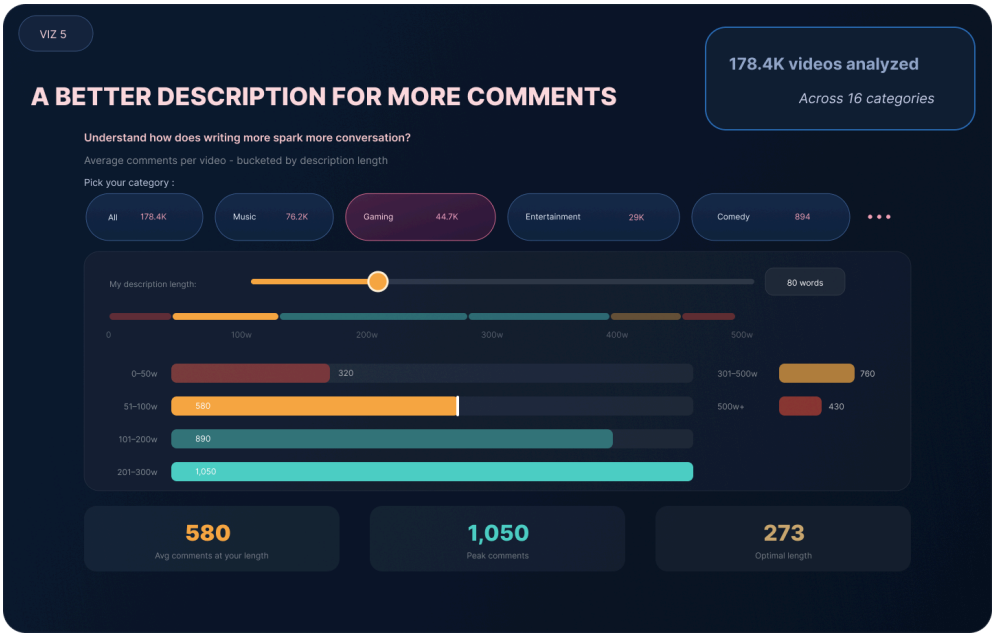
This visualization will be implemented using **D3.js with SVG**, as a bubble chart where bubble size represents tag frequency and color encodes average performance uplift. The layout will be generated with force simulation to avoid overlap and improve readability.

This visualization relies on the following lectures.

- D3.js
- Interaction, filtering, aggregation
- Perception, cognition, color
- Mark and channels

With the core visualizations defined, we now present additional ideas that could further enhance the project.

4 . Description Analysis & Prevision Chart



This visualization will be implemented using D3.js with SVG, as well as some feature extraction and bucketting on the pre-processing side. It would allow users to estimate the potential reception of a video given its category and the size of its description. As we don't currently know the correlation between a description size and its virality, it's currently an optional task.

This visualization relies on the following lectures:

- D3.js
- Data
- Interaction, filtering, aggregation
- Marks and channels
- Text

5 . Title Pattern X Category Analysis Grid

We also would like to be able to see, for any categories, if certain title patterns (full uppercase title, or clickbait keywords for example) often result in bigger view amounts. This could be achieved by a grid, with the different known title patterns on the Y axis, and the category on the X axis. If we somehow get access to the view evolution of any videos, rather than the final amount, we could make the following visualisation too :

This visualization relies on the following lectures:

- D3.js
- Data
- Interaction, filtering, aggregation
- Perception, cognition, color
- Marks and channels
- Text

6 . SubGenre Clustering

Another good idea would be to get the title + keywords of every videos, embed them using a known embedding model, and flatten them on a 2-dimensionnal space, to see if some clustering appears (which would effectively be sub-genre or “emerging” genre), and could also make certain trends more visible (specific subgenre being more trendy than others). As we don't currently know if such correlations exist, this visualisation will be optional.

This project explores how YouTube creators optimize metadata to improve video visibility, focusing on how performance is intentionally shaped rather than simply observed. Through a set of interactive visualizations, we highlight key factors such as publishing time, emoji usage, and tags, providing actionable insights for content creators. The core MVP ensures clear and interpretable results, while additional ideas offer opportunities for deeper analysis and extension.

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